

Smart Outcome Predictor

Comprehensive Machine Learning Pipeline & Predictive Performance Report

Dataset Size: 5,200 Records (19 Features)	Classification Target: completion_status	Regression Target: final_score
Pipeline Tasks: Preprocessing, Bagging, Boosting	Train/Test Split: 80% / 20% (4,160 / 1,040)	Date: September 2026

1. Executive Summary

This project presents an end-to-end Machine Learning solution designed to predict student performance and engagement in online learning environments. Utilizing a dataset of **5,200 students** across 19 behavioral and demographic variables, the pipeline concurrently addresses two critical tasks:

- **Binary Classification:** Predicting course completion status (completion_status: 0 or 1).
- **Continuous Regression:** Predicting the final numerical score achieved by the student (final_score: scale 0–100).

Through systematic experimentation across Decision Trees, Bagging Ensembles, AdaBoost, Gradient Boosting, LightGBM, and XGBoost, the best performance was obtained using **Gradient Boosting Classifier (74.13% Accuracy)** for outcome completion and **Gradient Boosting Regressor ($R^2 = 0.4855$, MAE = 7.85)** for final score estimation.

5,200 TOTAL STUDENT RECORDS	74.13% TOP CLASSIFICATION ACCURACY	0.4855 TOP REGRESSION R^2 SCORE	7.85 MIN MEAN ABSOLUTE ERROR
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2. Dataset Overview & Architecture

The dataset contains student demographic attributes, learning interaction metrics, and course specific meta-data. Features were engineered and transformed prior to modeling.

Feature Category Breakdown

Feature Type	Columns Included
Numerical Identifiers & Metrics	age, week_of_year, sessions, time_spent_hours, videos_watched, quiz_attempts, assignments_submi, attendance_rate
Categorical Features	country_region, device_type, education_background, course_level, course_category
Temporal Feature Engineering	course_start_date

3. Data Preprocessing & Pipeline Construction

To prevent data leakage, a strictly contained `ColumnTransformer` pipeline was implemented within an 80/20 stratified train-test split configuration.

- **Training Samples:** 4,160 instances
- **Testing Samples:** 1,040 instances
- **Post-Preprocessing Feature Dimension:** 30 transformed columns (including one-hot encoded categories).

Pipeline Architecture Note

The preprocessing pipeline imputes missing values in numerical columns (e.g., 112 missing values in `time_spent_hours`, 81 in `avg_quiz_score`, 80 in `attendance_rate`) using median strategy, while transforming categorical variables into binary indicator flags.

4. Model Benchmarking & Experimental Results

Part C: Decision Trees vs. Bagging Ensembles

Bagging (Bootstrap Aggregating) was evaluated with 50 base decision trees to measure variance reduction against standalone decision trees.

Classification Comparison

Model	Accuracy
Single Decision Tree (depth=6)	71.44%
Bagging Classifier (50 estimators)	73.37% +1.93%

Regression Comparison

Model	MAE	RMSE	R ² Score
Single Decision Tree	8.9819	11.3081	0.3156
Bagging Regressor	7.9749	9.9474	0.4704 +0.1548

Part D: Advanced Boosting Frameworks Comparison

Sequential boosting techniques were tested across identical evaluation sets. Boosting algorithms iteratively correct the residual errors of preceding weak learners.

Model Architecture	Classification Accuracy	Regression MAE	Regression RMSE	Regression R ²
AdaBoost	72.88%	8.7868	10.8309 (MSE 117.31)	0.3721
Gradient Boosting (Default lr=0.1, n=100)	73.65%	7.8476	9.8049 (MSE 96.14)	0.4855
Gradient Boosting (Tuned lr=0.05, n=100)	74.13% Best	--	--	--
LightGBM	72.69%	7.9512	9.9070 (MSE 98.15)	0.4747

Model Architecture	Classification Accuracy	Regression MAE	Regression RMSE	Regression R ²
XGBoost	72.31%	7.9838	9.9315 (MSE 98.64)	0.4721

5. Hyperparameter Tuning & Analysis

Hyperparameter exploration on Gradient Boosting Classifier demonstrated clear sensitivity to learning rate (`learning_rate`) and number of estimators (`n_estimators`):

- **Learning Rate Sensitivity (`n_estimators=100`):**

- `lr = 0.01`: 69.90% Accuracy (Underfitting)
- `lr = 0.05`: **74.13% Accuracy (Optimal)**
- `lr = 0.10`: 73.65% Accuracy
- `lr = 0.20`: 72.12% Accuracy (Overfitting)

- **Estimator Scale Sensitivity (`lr=0.10`):**

- `n = 25`: 73.94% Accuracy
- `n = 50`: 73.94% Accuracy
- `n = 100`: 73.65% Accuracy
- `n = 200`: 72.40% Accuracy

6. Key Technical Insights & Recommendations

1. **Ensembling Gains:** Transitioning from a single decision tree to Bagging or Gradient Boosting improved classification accuracy from 71.44% to ~74.13% and boosted variance explained in score prediction (R^2) from 31.56% to 48.55%.
2. **Top Performing Model:** Gradient Boosting provides the highest accuracy and lowest prediction error across both classification and regression tasks.
3. **Efficiency vs. Accuracy:** LightGBM exhibited extremely fast training speed (~0.39 to 0.41 seconds) while retaining strong predictive parity (72.69% accuracy, $R^2=0.4747$), making it ideal for real-time web applications or larger scaled datasets.